



Data Driven Education

Discussion Forum Subteam

Presentation 3 - 11/19/2025

Meet the Team



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Introduction

- Online Discussion platform is crucial in modern education and digital studies.
- **Ed Discussion and EdX:** widely used online discussion forum
 - EdX offers Massive Open Online Courses (MOOCs)
 - enables students and instructors to interact
 - discuss course materials asynchronously
- Goals
 - Studying unique characteristics of Online Discussion Forums
 - Identifying students' progress and struggles
 - Predicting students success from their engagements
 - Helping instructors visualize findings

Research Questions

- How to use data science skills to provide insights into student engagement and attitudes toward the course?
 - Sentiment analysis with cognitive presence
- How to use machine learning techniques to predict students' success in the course based on their discussion engagements?
 - Utilizing Gradient Boost
- What is the best way to present discussion analytics to instructors in an actionable and user-friendly dashboard?
 - React dashboard

Literature Review

Literature Review:

Goals:

- Understand how instructors interpret dashboard data to design clearer, more actionable visualizations.
- Use prior instructor-facing dashboard designs to shape features that turn sentiment and engagement data into meaningful teaching insights.
- Incorporate best practices for dashboard adoption by adding supportive elements like explanations and guided interpretations.

Body of Knowledge:

- “A Closer Look at Instructor Use and Sensemaking of Learning Analytics Dashboards.” (AlZoubi, D. et al. 2024)
 - Investigates how instructors actually interpret and act on dashboard indicators, and which visualizations/metrics lead to real pedagogical decisions.

Literature Takeaways:

“A Closer Look at Instructor Use and Sensemaking of Learning Analytics Dashboards.” (AlZoubi, D. et al. 2024)

- Highlights that instructors rely on sense-making processes (not just data consumption) when interpreting dashboard indicators.
- Shows that contextual information and explanatory cues improve instructors' ability to act on analytics insights.
- Demonstrates that dashboard effectiveness depends on alignment with instructors' pedagogical beliefs and teaching practices.
- Emphasizes the need for actionable, decision-support features rather than purely descriptive visualizations.
- Identifies scaffolding and guided interpretation as key factors in improving instructor adoption and meaningful use of dashboards.

Data Science

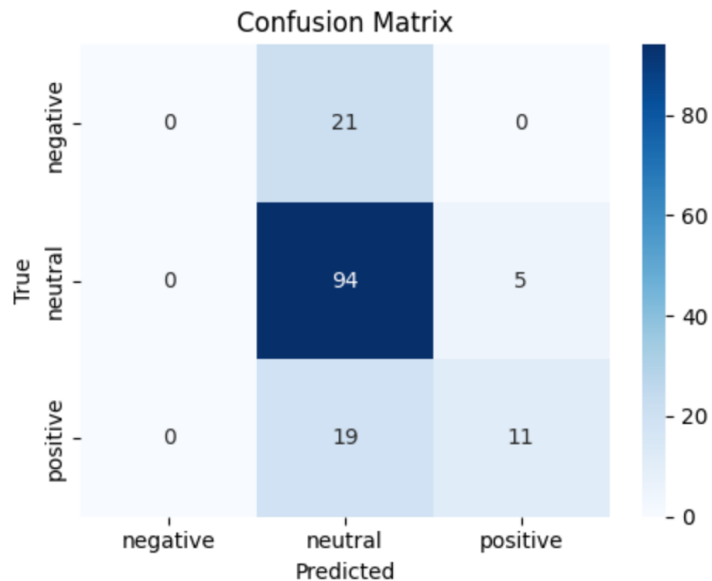
Current Progress

- Transitioned from BERT → DistilBERT sentiment analysis model
 - o Maintain 97% of BERT's language understanding capabilities while being 40% small and 60% faster
- Devised two parallel models
 - o Raw versus Clean Data
 - o Obtain results to be compared against machine learning data using confusion matrix

1272		negative	\x3564376 course-v1: i am sorry you are unhappy with the level of support but i have no way to automatically know th
1273		negative	\x3564373 course-v1: sorry i find it very frustrating that i have to manually follow every thread to know something ne
1274		neutral	\x3564373 course-v1: a few points i want to reemphasize 1 the activities and exercises are all newly redesigned for
1275		neutral	\x3564373 course-v1: i just checked it and it worked perfectly i have run into an issue with a firewall keeping clinfir
1276		neutral	\x3564366 course-v1: there have been some issues since clinfir was upgraded i will look into this but for now i hav
1277		neutral	\x3564333 course-v1: i posted an entirely new ndc activity a few days ago that does not involve your using any apis
1278		neutral	\x3564333 course-v1: i want to emphasize once again that the activities and exercises are mostly newly designed fo
1279		neutral	\x3564333 course-v1: as explained in the quiz instructions you have unlimited time but can answer each question o
1280		neutral	\x3564333 course-v1: providing an activity for cpt has always been a challenge because it is a forprofit product prod
1281		neutral	\x3564333 course-v1: anne you can email me at markbraunsteincgatechedu
1282		positive	\x3564323 course-v1: thanks has been updated it is a continuing challenge to keep up with changes in the web too

[339/339 02:14, Epoch 3/3]	
Step	Training Loss
10	1.051900
20	0.936100
30	0.864800
40	0.507100
50	0.765200
60	0.719100
70	0.787200
80	0.645600
90	0.568100
100	0.699600
110	0.688900
120	0.582700
130	0.481200
140	0.443700
150	0.404200
160	0.489700
170	0.300500

Current Progress, cont.



*Generated by running Distilled BERT model
on test dataset*

Statistics:

- Accuracy: 72.4%
- Precision:

Negative	Neutral	Positive
0%	70.1%	68.75%

- F1 Score:

Negative	Neutral	Positive
0%	80.6%	47.8%

- Insights:
 - Biased towards neutral predictions
 - Struggles to distinguish emotionally charged language from neutral tones

Current Progress, cont

Cognitive Presence - how students move through learning process approaching new problems, growing in understanding, conveying their learning

1. Triggering Event

- Confusion, misconceptions, deeper inquiry
- Associated with *negative* tone
- Model fails to identify these moments

2. Exploration

- Ask questions, share facts, brainstorm, exchange surface-level ideas
- Model can map to *neutral* tone with high confidence

3. Integration + 4. Resolution

- Breakthroughs, integration of ideas, confidence, successful problem-solving, “aha” moments
- Maps to positive tone
- Model has low positive sentiment prediction

Recap of the Semester

Fall 2025

- August – mid-October:
 - Continue exploring text-analysis models for **sentiment analysis**
 - **BERT** - Google's NLP model that can learn a wide breadth of tasks relating to language
- Mid-October – December:
 - Analyze dataset to understand student levels of **cognitive presence**
- December:
 - Using insights, revamp dashboard from last semesters

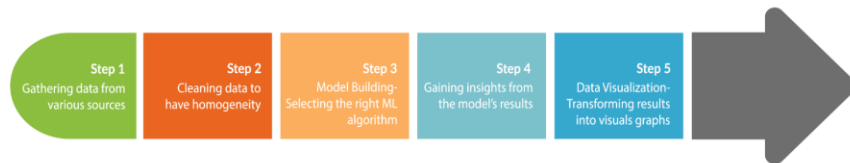
Future Goals

- Improve DistilBERT model's accuracy results by training it on a more balanced positive/neutral/negative edX discussion forums data
 - May need to leverage and improve Spring 2025 synthetic DF dataset
- Fine-tune current model to specifically identify cognitive presence phases in each comment
 - Reduction from standard sentiment analysis to cognitive presence detection
 - Retrain pre-trained sentiment analysis models on labeled dataset to fulfill more specific tasks
- Correlate cognitive presence identification with students' success in courses
 - Combined effort with ML subteam

Machine Learning

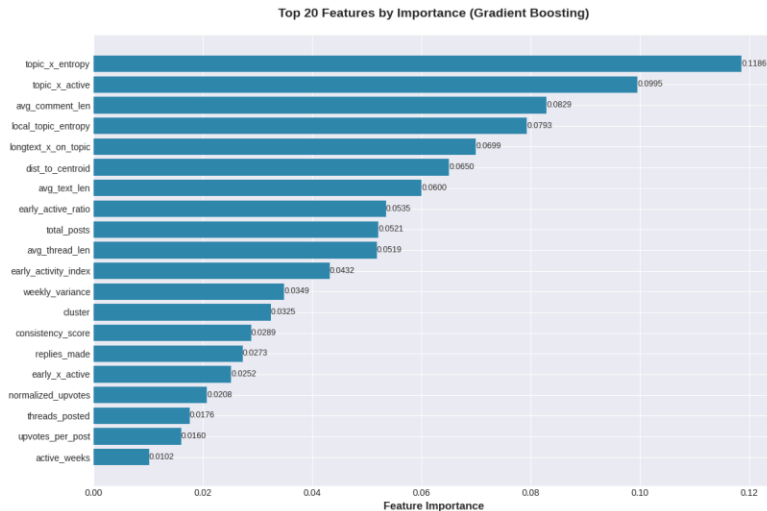
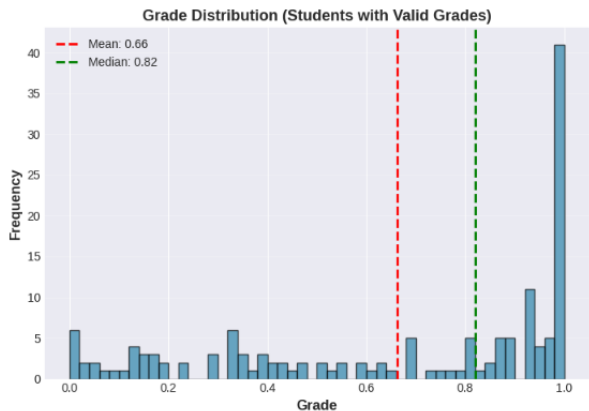
RECAP: Goals for the Semester

- Using data from CS1301 2017 & 2018
 - Join grades, forum post, and forum thread datasets with queries
 - Clean null/blank information
- Process data to create a bi-modal student behaviors model
 - Our target variable is student grades and we want to identify which forum behaviors predict academic success
 - We want to study 2 topics:
 - Text embeddings (semantics, etc.)
 - Time-series (frequency, timing in sem, etc.)
- Once identifying which behaviors are key after fusing our bi-modal model, we can create visualizations of those traits and their correlation to grades



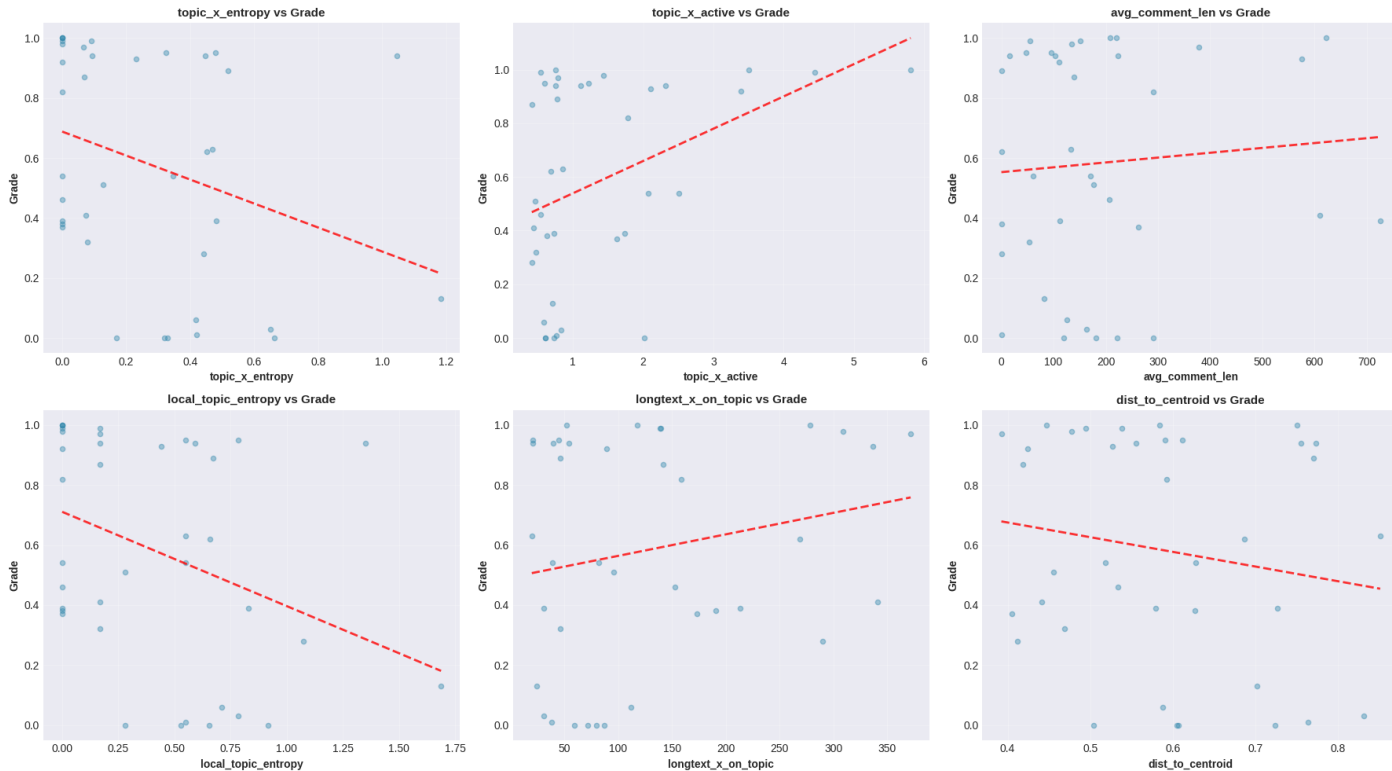
Improvements Made

- Switching from Ridge Regression -> Gradient Boosting
 - There was an improvement for R^2 from .086 to .153 on the training data
- Visualized lots of slices of this ML model: **grade distribution, feature importance**



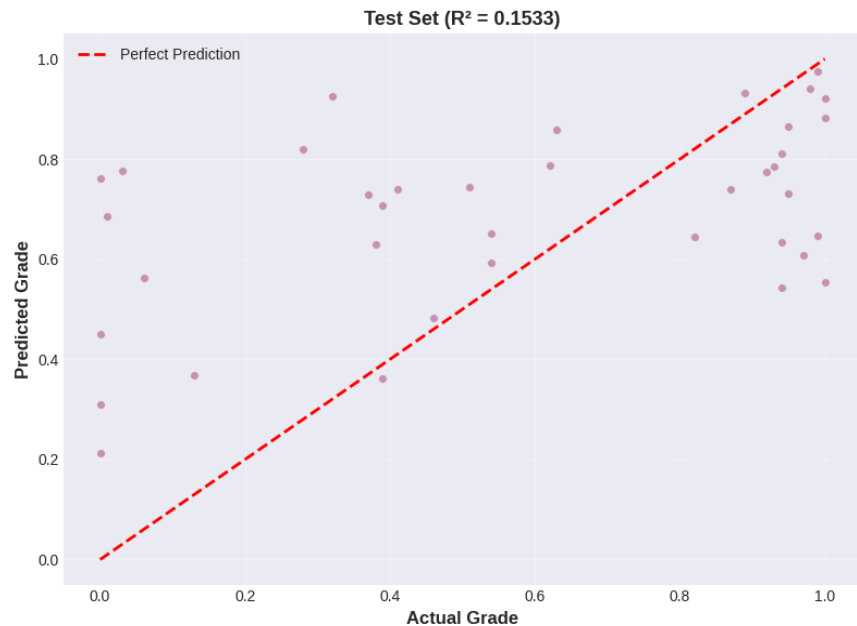
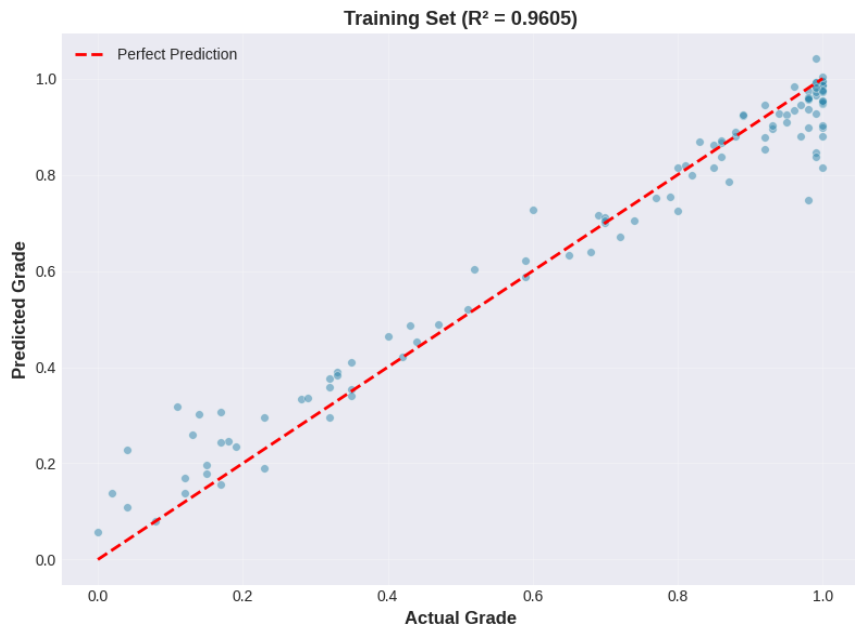
Additional Viz

- Visualized lots of slices of this ML model: line of best fit for top features



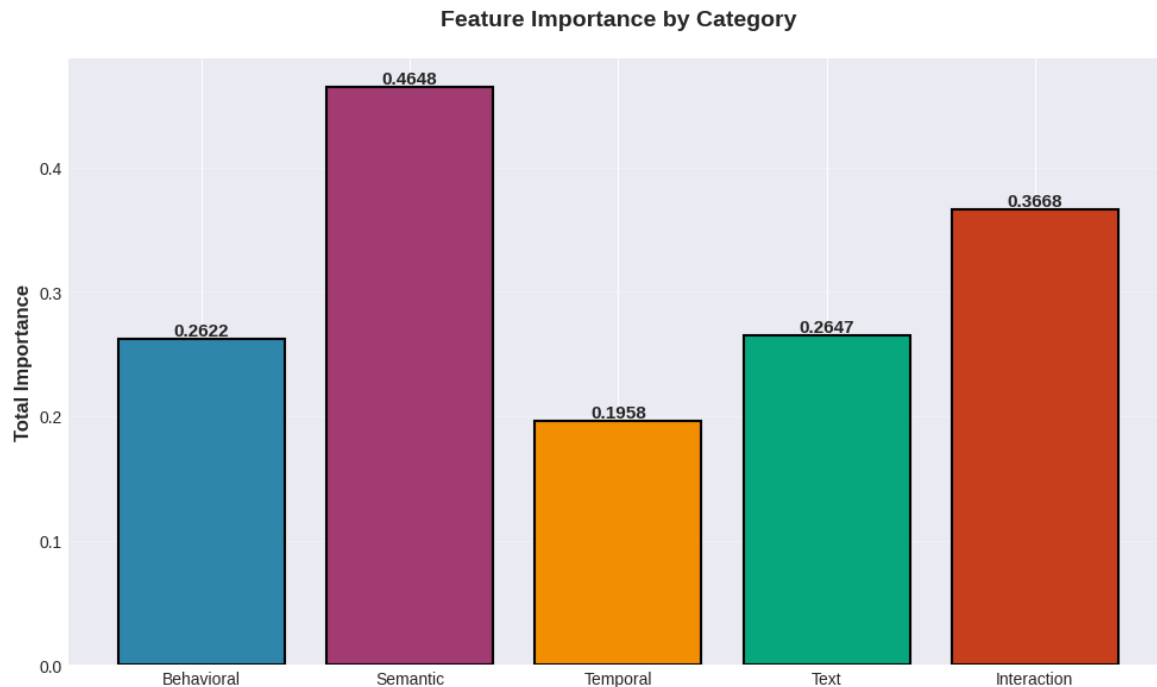
Additional Viz

- Visualized lots of slices of this ML model: **training vs test data**



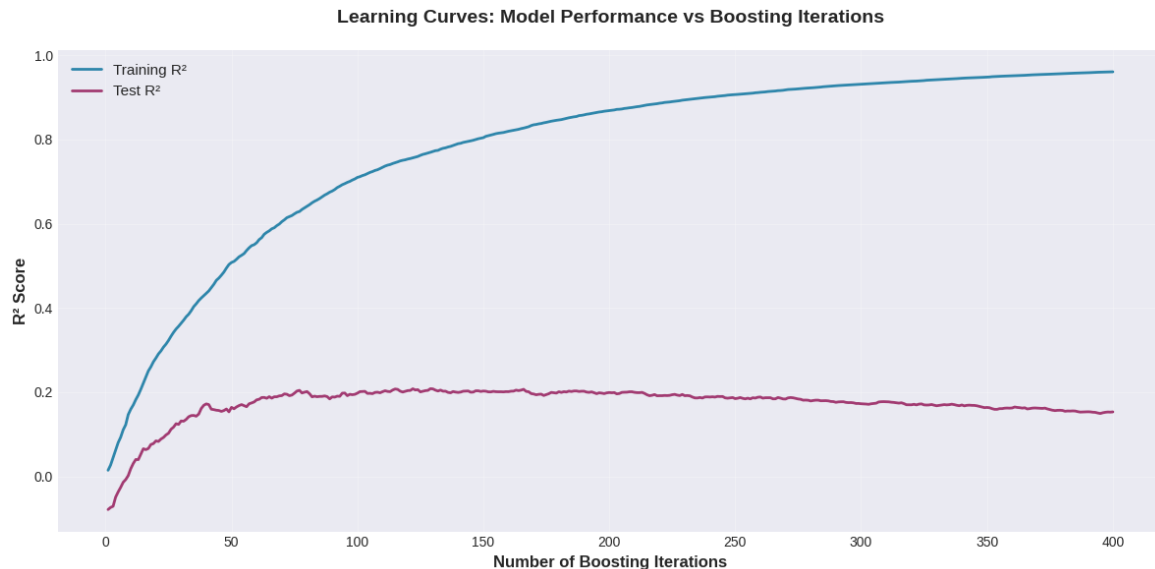
Additional Viz

- Visualized lots of slices of this ML model: **importance of multimodal features**



Future Goals

- The gradient boosting model is good and has improved BUT needs improvements
- We need to prevent the model's overfitting to training data
- This is a great foundational model and experiment to show there is more progress to make...



Dashboard

Sentiment Analysis Result

Example Results

Example 1

Sentence:

"I absolutely love the course."

Sentiment:

POSITIVE

Example 2

Sentence:

"I hate this"

Sentiment:

NEGATIVE

More examples from CS1301

Select a Sentence

Choose a sentence from the dropdown to see the result

Select a sentence...

Awesome and thanks you so much. It worked.

Thanks, I had figured out that part but I am sorry that I did not describe clearly. I knew that I was evaluating only the first line, but I think did not realize that how could I resolve that problem. May be by changing the indent of the return? But may be there is more to it as I have tried a lot of ways.

I have seen a book about python projects which may be fun on the side. It could be advanced though....

You're welcome!

The secret was just about using the "math.e". It's a mathematical constant.

Analysis Result

Selected Sentence:

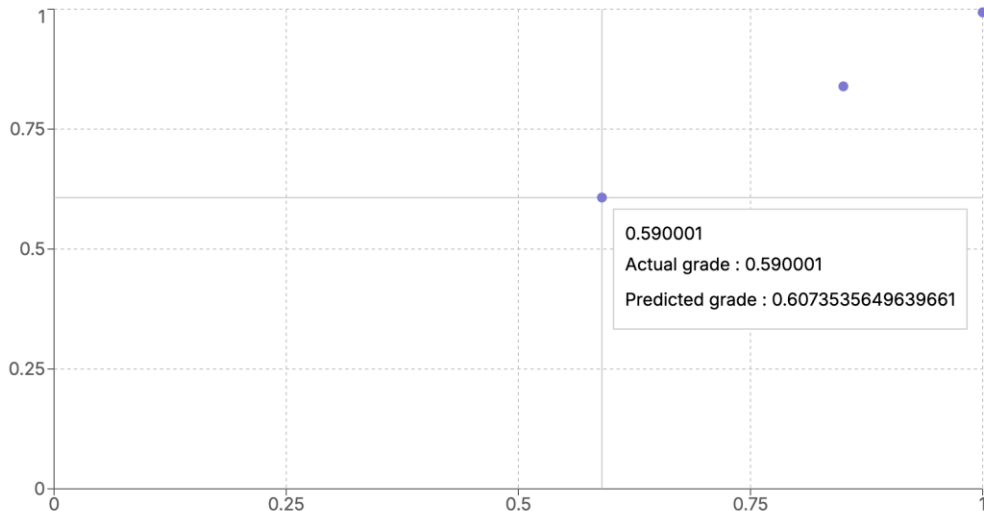
"Awesome and thanks you so much. It worked."

Sentiment:

POSITIVE

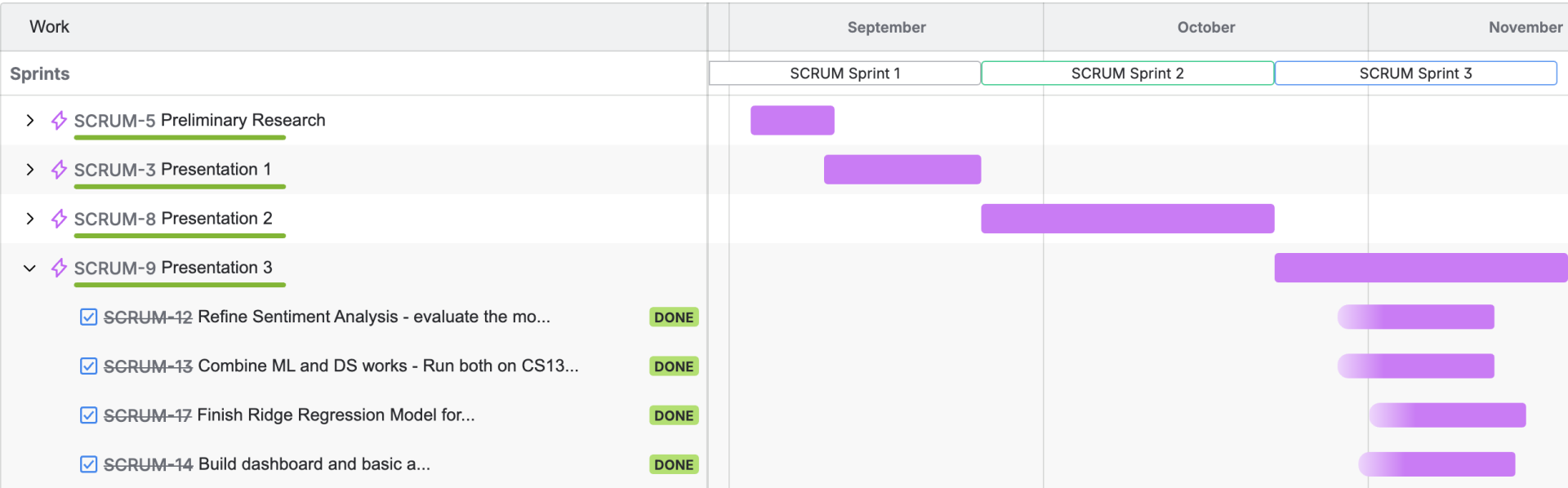
Show students'
sentiments in relation to their
grades

Actual vs. Predicted Grade



● Students

Project Timeline



Jira - For task tracking and management

Conclusion

- Achievements:

- Enhanced analysis of online discussion forum data, focusing on post contents and grades
- Explored sentiment analysis utilizing DistilBERT
- Identified key factors in relation to students' final grades with Gradient Boost
- Initialized React Dashboard visualizations

- Next Steps:

- Increase the data size -> improve sentiment analysis & gradient boost
- Making dashboard more interactive

- Ultimately:

- Improve course design and refine teaching strategies
- Personalize learning for foreseeable weakness
- Promote student's engagements and grades
- Ensure a more responsive and adaptive online learning experience

Citations

1. AlZoubi, D., and E. Baran. “A Closer Look at Instructor Use and Sensemaking Processes of Analytics Dashboards: Past, Present, and Future”. *Journal of Learning Analytics*, vol. 11, no. 2, Apr. 2024, pp. 1-22, doi:10.18608/jla.2024.7961.

Thanks for listening!

Q&A